

Module 4 Programming II Java Hamburger Franchise Order Management System Lab Assignment



Java Hamburger Franchise Order Management System Lab Assignment

Objective:

Design and implement a Java console-based application that interacts with a MariaDB database to receive and manage customer hamburger orders. The application should allow users to place orders, view order history, and perform basic order management tasks.

Prerequisites:

- Basic knowledge of Java programming.
- Understanding of JDBC for database connectivity.
- MariaDB installed and accessible.

Tasks:

1. Setup MariaDB Database (15 points):

- Create a MariaDB database named `hamburger_db`.
- Design a table named `orders` with columns: `order_id` (INT, AUTO_INCREMENT, PRIMARY KEY), `customer_name` (VARCHAR(255)), `burger_type` (VARCHAR(255)), and `order_date` (TIMESTAMP).

2. Implement Order Management System (60 points):

- Create a Java console application that provides the following functionality:
 - **Place Order (20 points):**
 - Accept customer name and burger type as input.
 - Insert a new order into the database with the current timestamp.
 - **View Order History (20 points):**

- Display a list of all orders from the database.
- Include order_id, customer_name, burger_type, and order_date.
- **Cancel Order (20 points):**
 - Allow the user to cancel an existing order by providing the order_id.

3. Exception Handling (10 points):

- Implement exception handling to address potential errors (e.g., database connection issues, user input validation).

4. Skeleton Code:

```
import java.sql.*;
import java.util.Scanner;

public class HamburgerOrderSystem {

    private static final String DB_URL = "jdbc:mariadb://localhost:3306/hamburger_db";
    private static final String DB_USER = "your_username";
    private static final String DB_PASSWORD = "your_password";

    public static void main(String[] args) {
        createTableIfNotExists();
        Scanner scanner = new Scanner(System.in);

        while (true) {
            displayMenu();
            int choice = scanner.nextInt();

            switch (choice) {
                case 1:
                    placeOrder(scanner);
                    break;
                case 2:
                    viewOrderHistory();
                    break;
                case 3:
                    cancelOrder(scanner);
                    break;
                case 4:
                    exitSystem();
                    break;
                default:
                    System.out.println("Invalid choice. Please try again.");
            }
        }
    }

    private static void displayMenu() {
        System.out.println("\nHamburger Order Management System");
    }
}
```

```
        System.out.println("1. Place Order");
        System.out.println("2. View Order History");
        System.out.println("3. Cancel Order");
        System.out.println("4. Exit");
    }

    private static void createTableIfNotExists() {
        // TODO: Implement table creation logic
    }

    private static void placeOrder(Scanner scanner) {
        // TODO: Implement order placement logic
    }

    private static void viewOrderHistory() {
        // TODO: Implement order history viewing logic
    }

    private static void cancelOrder(Scanner scanner) {
        // TODO: Implement order cancellation logic
    }

    private static void exitSystem() {
        System.out.println("Exiting the system. Thank you!");
        System.exit(0);
    }
}
```

Grading Rubric:

- **MariaDB Database Setup:**
 - Hamburger_db creation: 5 points
 - Orders table creation: 10 points

- **Order Management System:**

- Place Order: 20 points
- View Order History: 20 points
- Cancel Order: 20 points

- **Exception Handling:**

- Proper handling of potential errors: 10 points

- **Code Structure and Comments:**

- Clear and organized code: 10 points
- Comments explaining key sections: 5 points

Submission Guidelines:

- Submit a zip file containing the Java source code files (.java).
- Include a README file with any necessary instructions, especially MariaDB setup details.
- Ensure code readability and follow best practices.

Points 100

Submitting a file upload

Due	For	Available from	Until
Oct 8	Everyone	-	Oct 22 at 11:59pm